CODERED '26 | INDUSTRY CHALLENGE BRIEF

Problem Statement 4
HyperSpeed Quick Commerce

Disruptive logistics and ordering systems for minutes-level delivery at scale

OBJECTIVE Address the intense competition and logistical demands of the quick commerce sector by designing a disruptive logistical or ordering system that drastically increases delivery efficiency, operational speed and overall market competitiveness.

1. Challenge Overview

Company / Partner Name: MyStartupWave

Challenge Title: HyperSpeed Quick Commerce

Industry / Track: Logistics & Supply Chain / Quick Commerce

Challenge Contact: MyStartupWave Challenge Desk (mystartupwave@gmail.com)

Challenge Type: Product Innovation

Brief Summary: Quick commerce platforms compete on delivery promises of around 10 minutes, which depend on demand forecasting, dark-store operations, rider allocation and routing working together under tight margins. Teams will design and build a disruptive logistical or ordering system that measurably increases delivery efficiency and operational speed, validated through a realistic, reproducible simulation of orders, dark stores and riders.

2. Industry Context

Quick commerce has grown rapidly in Indian cities, with players such as Blinkit, Zepto and Swiggy Instamart promising groceries and essentials within minutes. The model relies on dense networks of dark stores, small high-velocity inventories and large rider fleets operating in congested traffic.

Margins are thin and every second counts: picking and packing, rider assignment, batching, last-mile routing and stock availability all directly affect delivery time, cost per order and customer retention. As players expand into new cities and categories, operational efficiency has become the key differentiator, and advances in forecasting, optimisation, real-time data and AI create room for genuinely new approaches.

3. Problem Statement

- **Problem and who is affected:** Quick commerce operators, dark-store managers and riders struggle to keep delivery fast and economical at scale; customers experience late deliveries, cancellations and out-of-stock items.
- **When and where it occurs:** Continuously during operation, and most severely at peak demand (meal times, evenings, weekends, festivals, rain), in congested urban areas, and when demand shifts unexpectedly.
- **Limitations of current solutions:** Rule-based rider assignment and nearest-store routing break down at peak; forecasts at dark-store and SKU level are noisy, causing stock-outs or wastage; picking layouts and processes are static; riders sit idle or are overloaded; and optimisations are often made in isolation rather than end to end.

- **Why it is worth solving:** Delivery-time promises are the core competitive claim of every quick commerce brand; stock-outs and delays directly drive cancellations and churn; and unit economics depend heavily on orders per rider-hour and store throughput, so even small efficiency gains create significant value.

4. Target Users and Stakeholders

Primary users: Quick commerce operations teams and dark-store managers.

Secondary users / stakeholders: Delivery riders, customers, inventory and supply-chain planners, and brands or suppliers that replenish dark stores.

Current user journey or workflow: Customer places an order → order is routed to the nearest dark store → a picker collects items → order is packed → a rider is assigned (often first-available) → rider navigates to the customer → rider returns to the store. Replenishment is usually planned from historical sales, often once a day.

Key user pain points: Long pick-and-pack times at peak, poor rider utilisation, cancellations caused by stock-outs, inaccurate ETAs, inefficient batching, and limited real-time visibility across stores and the fleet.

5. Product Opportunity

- **What users should accomplish:** Fulfil more orders within the promised delivery window, especially at peak and during disruptions.
- **Meaningful improvement:** Lower average order-to-door time and cost per delivery, with better rider utilisation and stock availability.
- **What makes it different:** A genuinely new idea in ordering, inventory positioning, dispatch, batching or routing, rather than an incremental tweak, with its impact proven quantitatively against a realistic baseline.

6. Expected Product Capabilities

Must-have capabilities:

- A clearly defined innovation in at least one of: demand forecasting and inventory placement, order batching, rider dispatch and routing, dark-store picking, or the customer ordering experience, integrated into an end-to-end order flow.
- A simulation environment modelling a city zone with at least 3 dark stores, 30+ riders, thousands of orders per day, realistic road travel times, demand peaks and SKU-level inventory.
- A baseline policy (e.g., nearest store + first-available rider) implemented in the same simulator, so improvements are measured on identical scenarios.
- A real-time operations dashboard showing orders, store load, rider positions and status, SLA breaches and key KPIs.
- Dynamic decision-making that adapts in real time to surges and disruptions (e.g., rain, rider shortage, stock-out at a store).
- Accurate delivery-ETA prediction shown to the customer at order time.

Nice-to-have capabilities:

- Predictive pre-positioning of inventory or riders ahead of forecast demand.
- Multi-store order splitting or substitution suggestions when items are out of stock.
- Fairness- and fatigue-aware rider assignment.
- A customer-facing ordering prototype with smart recommendations and delivery-slot nudges.
- Use of real open map data (e.g., the OpenStreetMap road network of an Indian city).

Expected user interactions: Operators monitor the live dashboard and intervene when alerts fire; the system automatically assigns, batches and routes orders; customers place orders and see accurate ETAs; riders receive optimised tasks.

Expected demonstration / deliverable: Run the baseline and the proposed system on the same simulated day, including at least one demand surge and one disruption; present the live dashboard; and show a side-by-side KPI comparison with a clear explanation of why the improvement occurs.

7. Technical and Operational Constraints

Technical or platform constraints: Any technology stack may be used. The decision engine must operate in real time within the simulation. The simulation must be reproducible (fixed random seeds) so that judges can rerun scenarios. Live data from any quick commerce company's private systems must not be used.

Performance, accuracy, or latency requirements: Dispatch and routing decisions within 2 seconds per order at a load of at least 500 concurrent active orders. Versus the baseline on the same scenario, the proposed system must achieve at least a 15% reduction in average delivery time or at least a 20% improvement in on-time (within-SLA) delivery rate, without increasing cost per order. ETA prediction error (MAE) must be under 2 minutes.

Security, privacy, or compliance requirements: Location and customer data must be synthetic or anonymised. Designs must respect road-safety and labour norms for riders.

Hardware, connectivity, or deployment limitations: Should run on a standard laptop or a free cloud tier; dashboards must run in a browser.

Budget or resource constraints: Use open-source optimisation and mapping tools where possible (e.g., Google OR-Tools, OSRM, OpenStreetMap). Any paid API usage must be documented.

Domain-specific safety requirements: Delivery-time improvements must come from better planning, not from assuming unsafe rider speeds. Simulated rider speeds must stay within realistic urban limits, and rider workload and break limits must be respected.

8. Available Resources and Support

Resources marked "Confirmed" are guaranteed by the partner. "Permitted" items may be used by teams at their own discretion and cost. "Under consideration" items are not yet confirmed.

APIs, SDKs, developer tools, or technology credits: Confirmed: none provided by the partner. Permitted: open-source tools such as OpenStreetMap, OSRM or GraphHopper, Google OR-Tools and SimPy, or other APIs available to the team.

Sample datasets or permitted data sources: Synthetic order, inventory and rider data generated by the team; public grocery-order or delivery datasets and open road networks, used within their licences.

Sandbox environments or test infrastructure: Not provided. Teams build their own simulation environment.

Documentation and reference materials: Google OR-Tools vehicle-routing documentation; OpenStreetMap and OSRM documentation; published research on dynamic vehicle routing and last-mile delivery.

Mentors, domain experts, or technical sessions: Confirmed: MyStartupWave engineers during the scheduled mentoring rounds (subject to the event schedule).

Other resources or support: Under consideration: a session with a quick commerce operations practitioner.

9. Business and Adoption Context

Intended customer or beneficiary: Quick commerce and hyperlocal delivery operators, dark-store franchisees, and logistics-technology providers.

Current alternatives or competing solutions: In-house dispatch and routing systems at large operators; third-party last-mile logistics platforms and route-optimisation SaaS products.

Potential value created: Shorter delivery times and lower cost per order, higher rider productivity, fewer cancellations and stock-outs, and better customer retention.

Potential adoption pathway: Pilot as a plug-in optimisation module for a single dark-store cluster, then expand into SaaS for mid-sized hyperlocal players without in-house optimisation teams.

Commercial or operational considerations: Integration with existing order-management systems and rider apps, real-time compute cost, and change management for store and rider operations.

Known assumptions or uncertainties: Simulated demand and traffic are representative enough to indicate real-world impact; operators would share data for a pilot.

10. Success Indicators

Functional outcomes: End-to-end simulation of ordering, picking, dispatch and delivery for both the baseline and the proposed system; a live dashboard; adaptive response to surges and disruptions.

User or customer outcomes: A higher share of orders delivered within the promised SLA; more accurate ETAs; better rider utilisation without unsafe workloads.

Technical performance indicators: ≥ 15% lower average delivery time or ≥ 20% higher on-time rate versus baseline; decisions in < 2 seconds at 500+ concurrent orders; ETA MAE < 2 minutes; reproducible results.

Business or operational indicators: Quantified impact on cost per order and orders per rider-hour, with a credible pilot plan.

Safety, reliability, or compliance indicators: No reliance on unsafe speeds; rider workload and break limits respected throughout the simulation.

11. Scope and Boundaries

In scope: Design of the logistics or ordering innovation, the simulation, baseline comparison, operations dashboard and ETA model.

Out of scope: Real-world deployment with live riders, payment systems, procurement contracts, a full customer app (optional only), and physical hardware such as drones or robots (concepts may be discussed but must be simulated).

Permitted assumptions or simplifications: A single city zone and a limited SKU catalogue (e.g., 200–500 SKUs) are sufficient. Travel times may be approximated from open road networks.

Dependencies (systems, approvals, resources): Open map data, optimisation libraries and compute for running the simulation.

12. Industry Engagement

Will the company provide mentors or subject-matter experts? Yes. Engineers from MyStartupWave will be available during the scheduled mentoring rounds for architecture, product and feasibility reviews.

Will teams receive feedback during development? Yes. Teams will receive structured feedback at the designated mid-event checkpoint(s), focused on scope, technical approach and progress against the success indicators.

Will the company participate in evaluation or demonstrations? Yes. MyStartupWave representatives will attend the final demonstrations and take part in evaluation alongside the CODERED '26 jury.

Is further validation, pilot discussion, or product exploration possible after the event? Possible. Teams with exceptional solutions may be invited to a follow-up discussion on further validation, mentoring or pilot exploration.

Any follow-on engagement is subject to mutual interest and separate agreement.

13. Intellectual Property and Participation Terms

The following terms apply to all participating teams and are communicated before development begins. Participation does not imply any automatic transfer of ownership or commercial rights; any subsequent arrangement must be agreed separately.

Data and resource usage conditions: Use only synthetic data, openly licensed datasets and open map data. Do not scrape or reverse-engineer any quick commerce company's apps, APIs or private data.

Confidentiality requirements, if any: None. This brief and any materials shared for this challenge are non-confidential.

Intellectual property and ownership terms: Teams retain full ownership of the code, designs, models and other intellectual property they create. Participation does not transfer any ownership or commercial rights to MyStartupWave.

Restrictions on using company names, data, or materials: Do not use MyStartupWave's name or branding without written permission. Names of existing quick commerce companies may be referenced only descriptively, for market context.

Terms governing subsequent pilots, licensing, investment, employment, or commercial collaboration: Any pilot, licensing, investment, employment or commercial collaboration after the event is subject to mutual interest and a separate written agreement.

14. Partner Contact and Clarifications

Primary contact: MyStartupWave Challenge Desk

Email: mystartupwave@gmail.com

Reference documentation / resource links: This challenge brief; Google OR-Tools documentation; OpenStreetMap and OSRM documentation.

Clarification process: Send questions in writing to the challenge contact or raise them through the official CODERED '26 helpdesk. Answers that affect scope or evaluation will be shared with all teams on this challenge at the same time, to keep the competition fair.

Submission Checklist

- ☑ The problem is real, relevant, and clearly stated.
- ☑ Target users and desired outcomes are identified.
- ☑ Teams have freedom to develop their own solutions.
- ☑ A working prototype is reasonably achievable within the challenge timeline.
- ☑ Business value and adoption can be evaluated.
- ☑ Technical constraints and available resources are stated.
- ☑ Data, safety, confidentiality, and IP terms are clear.
- ☑ Evaluation expectations are communicated consistently.

THE GUIDING PRINCIPLE

The company defines the opportunity. The team defines the solution.